

Session 14:

- 1.) In Power BI Desktop, create a calculated column named 'OrderAmountINR' in a Sales table that multiplies the 'OrderAmount' column by 83 to convert USD to INR.

ANSWERS:

A **calculated column** is created in Power BI Desktop to convert the order amount from USD to INR.

The new column is named **OrderAmountINR**, and it is created using the following DAX formula:

$\text{OrderAmountINR} = \text{Sales}[\text{OrderAmount}] * 83$

This formula multiplies the **OrderAmount** column by **83** (conversion rate) to convert the values into Indian Rupees.

After applying this column, the table now shows both the original amount in USD and the converted amount in INR, making it easier to analyze sales data in local currency.

Overall, this calculated column helps in better understanding and comparing order values in INR.

OrderAmount	orderAmtINR
294.76	24465.08
105.43	8750.69
482.97	40086.51
462.75	38408.25
238.9	19828.7
335.1	27813.3
115.12	9554.96
118.63	9846.29
151.38	12564.54
349.29	28991.07
114.06	9466.98
485.84	40324.72
44.47	3691.01
104.71	8690.93
53.54	4443.82
387.25	32141.75
189.59	15735.97
241.12	20012.96
169.75	14089.25
322.34	26754.22

2.) Build a DAX measure called 'Total Likes' that sums up the 'Likes' column in a Social Posts table, similar to how Instagram shows total likes for a user.

ANSWERS:

A **DAX measure** is created in Power BI to calculate the total number of likes from the Social Posts table.

The measure is named **Total Likes**, and the following formula is used:

Total Likes = SUM('Social Posts'[Likes])

This measure adds all the values from the **Likes** column to give the overall total likes, similar to how platforms like Instagram show total engagement.

After creating the measure:

- Add a **table or card visual**
- Place **Post_Type** (optional) in rows
- Place **Total Likes (measure)** in values

This will display total likes for each post type and also the overall total.

Overall, this measure helps to easily understand **engagement performance across different types of posts**.

Post_Type	TotalLikes
Carousel	21097
Image	19233
Reel	6559
Video	13994
Total	60883

- 3.) Create a slicer for 'Genre' in a Movies table and use a DAX measure to display the average 'UserRating' for the selected genre, like filtering movies by category on BookMyShow.

ANSWERS:



A **slicer** is created in Power BI using the **Genre** field from the Movies table. This allows users to filter movies based on selected categories like Action, Comedy, Drama, etc.

A **DAX measure** is created to calculate the average user rating:

Average User Rating = AVERAGE(Movies[UserRating])

After creating the measure:

- Add a **Slicer visual** and drag **Genre** into it
- Add a **Card or Table visual**
- Place **Average User Rating (measure)** in values

Now, when a user selects any genre in the slicer, the measure automatically updates and shows the **average rating for that selected genre**, similar to how apps like BookMyShow filter movies by category.

This makes the report interactive and helps in quickly analyzing ratings based on user-selected genres.

- 4.) Compare calculated columns and measures by creating both: a calculated column 'DiscountedPrice' (Price - Discount) and a measure 'TotalDiscountedSales' (sum of DiscountedPrice) in a Flipkart-style Products table.

Hint: Observe how each updates when you use filters or slicers.

ANSWERS:

In Power BI, both a **calculated column** and a **measure** are created to understand the difference between row-level calculation and aggregated results.

First, a calculated column named **DiscountedPrice** is created using the formula:

DiscountedPrice = Products[Price] - Products[Discount]

This column calculates the discounted price for each individual product row. The value is stored in the table and remains fixed for each row.

Next, a measure named **TotalDiscountedSales** is created using:

TotalDiscountedSales = SUM(Products[DiscountedPrice])

This measure calculates the total of discounted prices dynamically based on the current filter context.

When filters or slicers (like Category) are applied, the **calculated column does not change** because it is pre-computed, but the **measure updates automatically** to show the total only for the selected data.

This comparison clearly shows that calculated columns work at the **row level**, while measures work at the **aggregate level** and respond dynamically to filters in the report.

Product_Name	Sum of Price	Sum of DiscountedPrice	Total_Discounted_Sales
Adidas Ultraboost	2204	1,759.66	1,759.66
American Tourister Backpack	50115	46,424.53	46,424.53
Casio Edifice Watch	60717	47,913.70	47,913.70
Levis Denim Jacket	1927	1,416.72	1,416.72
Mamaearth Onion Shampoo	868	812.67	812.67
Nike Air Max	559	526.05	526.05
Nivea Men Facewash	3261	2,533.13	2,533.13
Zara Slim Fit Shirt	3735	3,301.25	3,301.25
Total	123386	1,04,687.71	1,04,687.71

Category

- ☐ Select all
- ☒ Accessories
- ☒ Apparel
- ☒ Beauty & Grooming
- ☐ Electronics
- ☐ Health & Fitness

5.) List all the DAX data types you used in the above tasks and give one example of a real-world app field (from Zomato, Paytm, etc.) that would use each type.

ANSWERS:

Product_Name	AvgQty	Sum of DiscountedPrice	Total_Discounted_Sales
('iPhone 15 Pro', 'Electronics')	199.00	77,278.60	77,278.60
Boat Nirvanaa	118.00	66,505.49	66,505.49
Total	136.88	3,18,122.70	3,18,122.70

Discount

1,592.60 9,424.06

Answer (Short):

Different DAX data types used are:

- **Whole Number** → e.g., OrderCount (Zomato orders)
- **Decimal Number** → e.g., Delivery Time (Swiggy minutes)
- **Currency** → e.g., TotalAmount (Paytm ₹ value)
- **Text** → e.g., Product_Name / Restaurant Name
- **Date/Time** → e.g., OrderDate (Flipkart orders)
- **Boolean** → e.g., IsDelivered (True/False)
- **Blank** → Missing values (e.g., no discount)

These types help in proper calculation and data handling in Power BI.